# Reza Shahriarirad, M.D.

Physician-scientist in surgical innovation, artificial intelligence, and computer vision

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## EDUCATION

**Shiraz University of Medical Sciences** — *Doctor of Medicine (M.D.)*
2013 – 2020

## PROFESSIONAL EXPERIENCE

**Department of Surgery – Surgical Innovation, Mayo Clinic** — *Research Fellow*
2024 – Present · Rochester, MN

- Involved in about 40 surgical research projects, including 30+ IRB-approved studies and several industry collaborations across thoracic, bariatric, plastic and reconstructive, and abdominal-wall surgery.
- Works across the study lifecycle: design, IRB preparation, data curation, statistical analysis, manuscript writing, and patient recruitment.
- Builds AI-ECG and machine-learning models for surgical risk prediction and the Mayo Obesity Staging Index (MOSI) for bariatric decision support.

**Mayo Clinic** — *Research Affiliation – A-Star Lab*
2025 – Present · Rochester, MN, USA

- Surgical AI and computer vision research; teaching assistant for the computer vision session of the ACS Clinical Congress 2026 hands-on AI course.

**Thoracic & Vascular Surgery Research Center, Shiraz University of Medical Sciences** — *Senior Researcher & Project Manager*
2018 – 2024 · Shiraz, Iran

- Managed clinical and translational research projects at the Thoracic and Vascular Surgery Research Center.
- Founded and directed a research mentoring program; mentored 15 medical students who contributed to 50+ peer-reviewed publications.

**Khatam-al-Anbiya Hospital** — *General Practitioner — Emergency Department*
2020 – 2022 · Bushehr, Iran

## SURGICAL INNOVATION PROJECTS · MAYO CLINIC

- **ApoQlar × Apple Vision Pro** — Real-time stereoscopic 3D visualization for robotic surgery
- **GoNoGoNET** — AI-assisted surgical video analysis and safety-zone identification
- **ALARM / OmniWatch** — Wearable-based postoperative monitoring
- **INTERCEPTION & Alpenglow** — Robotic bronchoscopy, biopsy, and molecular sampling
- **CTA Perforator** — Computer vision-based reconstructive surgical planning
- **JelloX** — 3D digital pathology
- **Robotic surgery curriculum** — Curriculum development to improve resident training

## SELECTED PUBLICATIONS

**Metabolic and Bariatric Surgery and Change in the AI-ECG–Derived Biological-Age Gap: A Longitudinal Cohort Study**
Obesity Surgery · 2026 · First author

**Prediction of Sentinel Lymph Node Metastasis in Breast Cancer Patients Based on Preoperative Features: A Deep Machine Learning Approach**
Scientific Reports · 2024 · First author

**Global Trends and Research Landscape of Robotic Surgery in Plastic and Reconstructive Surgery: A Bibliometric Analysis (1999–2025)**
Journal of Robotic Surgery · 2026 · Co-first author

197 peer-reviewed publications in total. Full list: Google Scholar.

## RESEARCH IMPACT

**197** peer-reviewed publications (44 first/co-first, 48 senior)

**3,604** citations · h-index 25

**149** peer reviews for 67 journals

**2** issued patents

Google Scholar, October 27, 2026

## SKILLS

- Study design
- RCT coordination
- Data management
- Statistical analysis
- Systematic review and meta-analysis
- Clinical research coordination
- IRB preparation
- Machine learning and predictive modeling
- Scientific writing and editing

## TOOLS

SPSS · Python · MATLAB · NCSS · EndNote

## PATENTS

**A Machine Learning-Based System for Detecting Leishmaniasis in Microscopic Images**
Apr 2021 · Patent IR 140050140003000416

**Utilization of Chest Tube in Pediatric Caustic Injuries: A New Method for Esophageal Stenting**
Oct 2020 · Patent IR 139950140003006653

## HONORS

**Team Collaboration Award (Project Lead)**
Mayo Research Fellows' Association · Sep 2026

**World's Top 2% Scientists**
Elsevier Data Repository / Stanford · Dec 2023

**National Outstanding Student Researcher**
Iran Ministry of Health and Medical Education · Oct 2020

## EDITORIAL

**Guest Editor**
PLOS One · June 2022

**Associate Editor**
Shiraz E-Medical Journal · August 2021

## LANGUAGES

English (Fluent) · Farsi (Native)